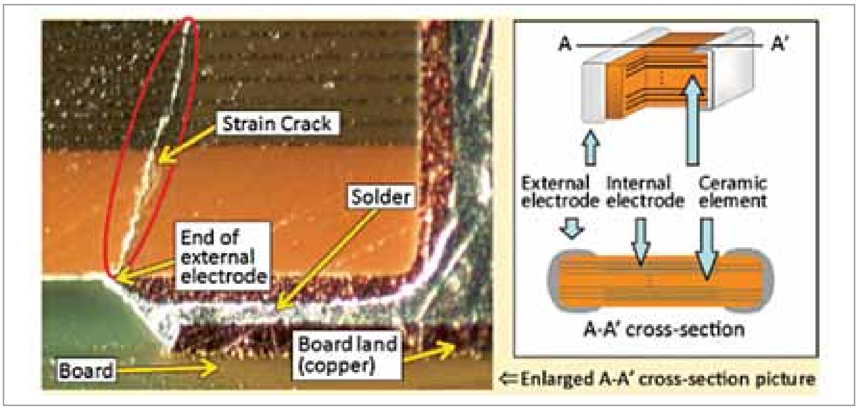

Fig. 1: A typical strain crack (Credit: Murata Manufacturing)

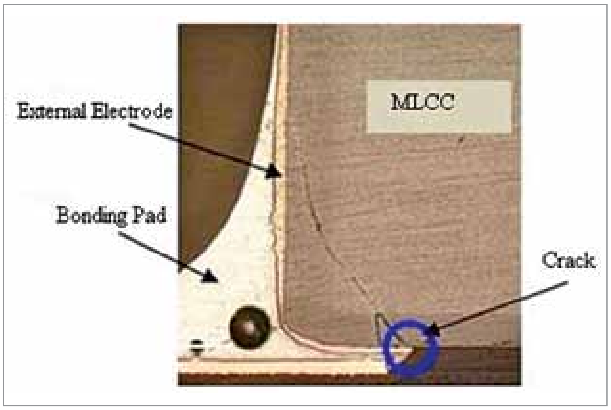

Fig. 2: MLCC crack due to vibration (Credit: AIRI Tech)

Fig. 3: A taller base with dummy terminals (Credit: Kemet Engineering Center)

cycling. Factors like vibrations, shock, and stress have clear effects on the passive components present in the system. Therefore, automotive engineers must look at the effects this will have on component reliability.

ESR: An important parameter

A key performance parameter for any DC-Link capacitor will be its equivalent series resistance (ESR) value. The lower the ESR of a capacitor, the greater its ability to smooth out high-frequency voltage disturbances passing through the powertrain. If the ESR values that these capacitors present within the electrical circuitry increase due to some form of damage, the whole powertrain operation could potentially be jeopardised.

While selecting capacitors that conform to these factors, vehicle manufacturers must keep the overall bill-of-materials expense as low as possible, so the chosen capacitors have to be cost-effective. There is, therefore, a constant tradeoff between efficiency and cost, and engineers need to maintain both sides of the equation for the vehicle to be optimal.

Effect of vehicle vibrations and stress

The adoption of high voltage levels in EVs and the location of motors away from the protection of the engine compartment in a few hybrid systems will have implications for the DC-Link capacitors incorporated into EV and hybrid powertrains. Situated between the battery and the traction inverter, these passive components are used to smooth out the voltage and prevent unwanted transient spikes from occurring. If they are to be effective, a high degree of robustness is needed. This will allow them to cope with the challenging application environment that automotive deployment represents.

Standard multilayer ceramic capacitors (MLCCs) are not well suited to automotive usage, as they are not able to cope with the acute mechanical stresses involved. If the PCB on which they are mounted flexes, due to either vibration or thermal cycling, the solder joints will be placed under considerable strain. This can lead to cracks appearing within the ceramic substrate. These cracks can result in moisture ingress arising which will, in turn, impact the capacitor performance. If these components are located close to the edge of the PCB, the stresses that they are subjected to are likely to be even greater.

Important considerations for engineers

- Low ESR value
- High operational temperature range
- Resilience to ongoing vibrations
- High degrees of robustness
- Small physical dimensions

Combating vibration-related issues

The vibrations witnessed in an automobile can come from a variety of sources and be at a wide range of different frequencies. Though these issues are intrinsic in automobile design, they can be mitigated in a number of ways as follows:

Specify polymer capacitors instead of MLCCs. Unlike normal